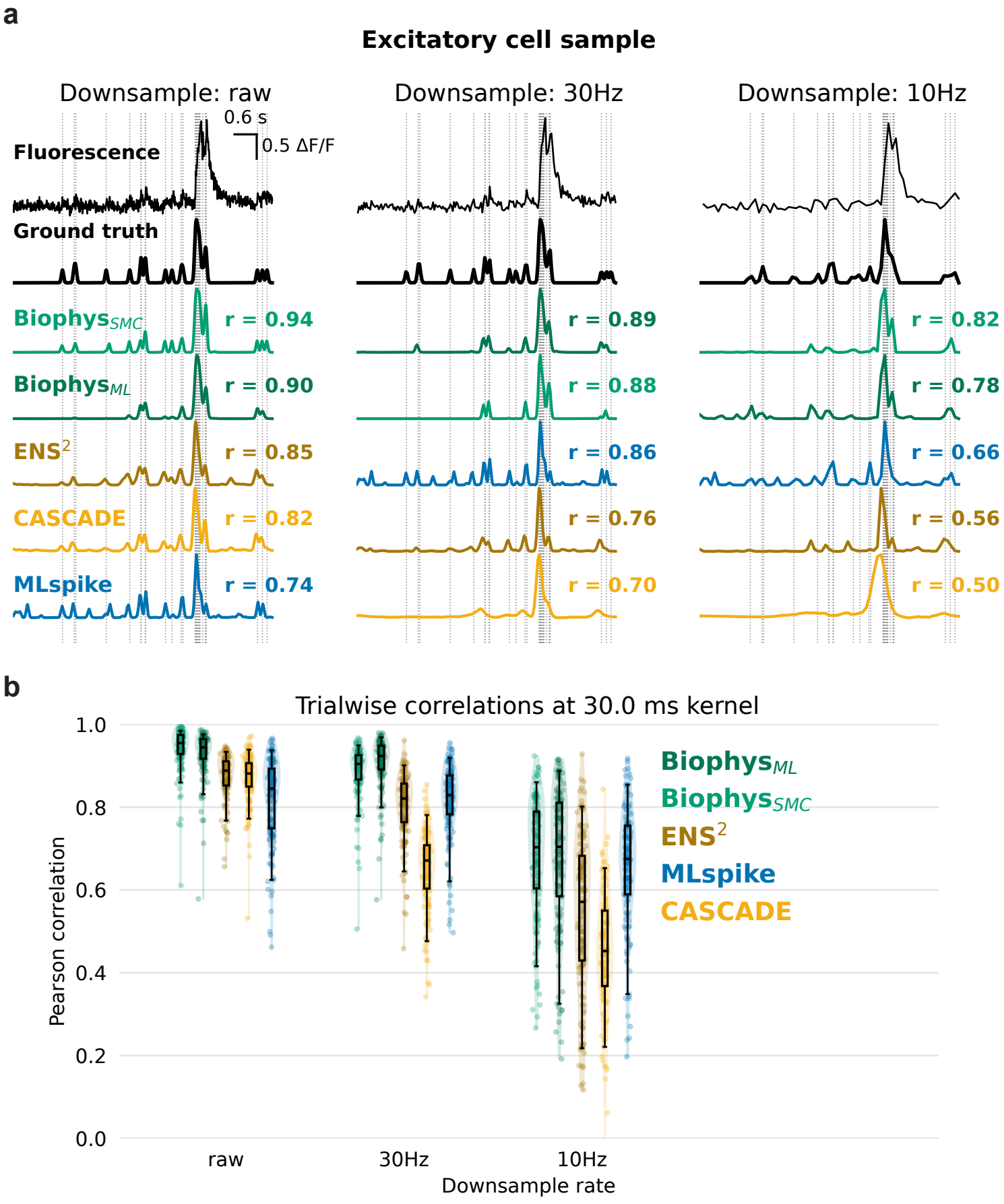

a
Excitatory cell sample
Downsample: raw
Downsample: 30Hz
Downsample: 10Hz
0.6 s
0.5 ΔF/F
Fluorescence
Ground truth
Biophys$_{SMC}$
Biophys$_{ML}$
ENS^2
CASCADE
MLspike
r = 0.94
r = 0.90
r = 0.85
r = 0.82
r = 0.74
r = 0.89
r = 0.88
r = 0.86
r = 0.76
r = 0.70
r = 0.82
r = 0.78
r = 0.66
r = 0.56
r = 0.50
b
Trialwise correlations at 30.0 ms kernel
Pearson correlation
1.0
0.8
0.6
0.4
0.2
0.0
raw
30Hz
10Hz
Downsample rate
Biophys$_{ML}$
Biophys$_{SMC}$
ENS^2
MLspike
CASCADE